

Element C

Presentation and Justification of Solution Design Requirements

Determination of Design Requirements

Through research, limitations put through by our school, and our interview with the Komodo dragon handlers, Cayden and James, of the Nashville Zoo we have outlined the following list of criteria and constraints.

*High Priority Requirements - Elements that are essential to the design based on criteria and interview with the stakeholder (ranked from highest priority to lowest).

Criteria

- *Height — 4-5 feet above the ground. The zookeepers established that they wanted the main part of the apparatus above the reach of the komodo dragons, which is around 4-5 feet. This is so that they are not able to reach, and therefore damage, the enrichment toy. They explained how they wanted the mounting of the enrichment toy to be above reach of the komodo dragons while there was a bottom feeder part that was in reach of the komodo dragons.
- *Food is detachable from the apparatus. The zookeepers expressed how if the food on the enrichment toy is unable to detach from the mounting part of the apparatus, the komodo dragons would attempt to eat the rest of the enrichment toy, which could cause trouble for both the komodo dragons and the medical team that has to remove the hardware from their stomachs.
- *The apparatus must reach a distance of at least 88ft. The zookeepers expressed how they wanted the enrichment toy to span across the length of their enclosure. Because the

enclosure is about 88 feet long from the points specified, the enrichment toy must be at least this long.

- *Can move equal to or greater than 13 mph. The zookeepers explained how they wanted the komodo dragons to be able chase the enrichment toy across the enclosure. Because the top speed of the komodo dragons is 13 miles per hour, the enrichment toy must be able to move this speed (the enrichment toy meaning the main system of the enrichment toy. The “zipline” that the toy moves across should be stationary).
- *Can hold 50 pounds of food. Lil Sebastian, the largest komodo dragon at the zoo, eats 50 pounds of food in one sitting. Thus, the enrichment toy should be able to carry 50 pounds of food. This is not of utmost priority because if the enrichment toy is not able to hold 50 pounds of food in one sitting, the zookeepers can go through multiple rounds.
- Must be able to withstand water and humidity. The natural habitat is very hot and humid, so when the komodo dragons are outside and using this toy, it must be usable in the humid conditions. Furthermore, the enrichment toy should be waterproof in the case that it is brought into the water by the komodo dragons. However, this is not a very high priority criteria because it is not very likely, and most materials that are strong enough for the komodo dragons to tug at can withstand the humidity and water.

Google Earth Distance Calculations



88 ft

Constraints:

- The komodo dragon enrichment toy must be completed by April 28th 2026
- The toy must cost less than \$250 to create.
- The communication is limited to some email after the initial meeting.
- There will be some member availability issues the week before the project is due.